

CHM 2045C: Exam 4 Comprehensive Study Guide

Quantum Mechanics, Periodicity, and Chemical Bonding

Spring 2026

Exam Strategy

Total Questions: 20 — **Math:** 3 — **Conceptual/Numerical:** 17

This exam focuses heavily on *why* atoms behave the way they do based on electron behavior. Master the trends and Lewis rules to secure the majority of the points.

1 Chapter 8: The Quantum Mechanical Model

1.1 8.1–8.2: Nature of Light

Concept: Light behaves as both a **wave** (diffraction/interference) and a **particle** (photoelectric effect). Particles of light are called *photons*.

- **Wave Formula:** $c = \lambda\nu$ (where $c = 3.00 \times 10^8$ m/s, λ is wavelength in meters, ν is frequency in s^{-1}).
- **Particle Formula:** $E = h\nu = \frac{hc}{\lambda}$ (where $h = 6.626 \times 10^{-34}$ J·s).
- **Relationship:** Short wavelength (λ) = High frequency (ν) = High energy (E).

1.2 8.4: Wave Properties of Matter (The de Broglie Wavelength)

Concept: If light can act like a particle, matter (like electrons) can act like a wave.

Mathematical Method: de Broglie Calculation

Formula: $\lambda = \frac{h}{mv}$

Critical Units:

1. Mass (m) **MUST** be in **kilograms** (kg).
2. Velocity (v) **MUST** be in **meters per second** (m/s).
3. λ will result in meters (m).

1.3 8.5: Quantum Numbers

Concept: Four numbers describe the "address" and properties of an electron in an orbital.

1. **Principal (n):** Energy level/Size. $n = 1, 2, 3, \dots$
2. **Angular Momentum (l):** Shape of the orbital. l ranges from 0 to $(n - 1)$.
 - $l = 0 \rightarrow s$ (sphere); $l = 1 \rightarrow p$ (dumbbell); $l = 2 \rightarrow d$ (clover); $l = 3 \rightarrow f$.
3. **Magnetic (m_l):** Orientation in space. m_l ranges from $-l$ to $+l$ (including 0).
4. **Spin (m_s):** Direction of electron spin. Either $+1/2$ or $-1/2$.

1.4 8.5: Hydrogen Electron Transitions (Bohr Model)

Mathematical Method: Calculating energy change when an electron jumps levels.

$$\Delta E = -2.18 \times 10^{-18} \text{ J} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

- **Absorption:** $n_f > n_i$ (ΔE is positive).
- **Emission:** $n_f < n_i$ (ΔE is negative; light is released).

2 Chapter 9: Periodic Properties

2.1 9.3: Rules for Electron Configuration

- **Pauli Exclusion Principle:** No two electrons in an atom can have the same four quantum numbers (max 2 electrons per orbital, must have opposite spins).
- **Hund's Rule:** When filling degenerate orbitals (same energy), electrons fill them singly first with parallel spins to minimize repulsion.
- **Aufbau Principle:** Electrons fill the lowest energy orbitals first ($1s \rightarrow 2s \rightarrow 2p \dots$).

2.2 9.6–9.8: Periodic Trends

Underlying Concept: Trends are driven by **Effective Nuclear Charge** (Z_{eff}), which is the net positive charge "felt" by valence electrons after accounting for shielding by core electrons.

Visualizing Trends (Arrows point toward INCREASE)

Trend	Horizontal ($\leftrightarrow$)	Vertical ($\updownarrow$)
Atomic Radius	$\leftarrow$ (Increase Left)	$\downarrow$ (Increase Down)
Ionization Energy (IE)	$\rightarrow$ (Increase Right)	$\uparrow$ (Increase Up)
Metallic Character	$\leftarrow$ (Increase Left)	$\downarrow$ (Increase Down)
Electron Affinity	$\rightarrow$ (More Negative Right)	$\uparrow$ (More Negative Up)
Effective Nuclear Charge	$\rightarrow$ (Increase Right)	(Slight Increase Down)

2.3 9.7: Magnetic Properties

- **Paramagnetic:** Atom or ion has **unpaired** electrons. Attracted to magnetic fields.
- **Diamagnetic:** All electrons are **paired**. Slightly repelled by magnetic fields.

3 Chapter 10: Chemical Bonding I

3.1 10.5–10.6: Bond Classification

Determined by the difference in electronegativity (ΔEN):

- **0.0 – 0.4:** Pure (Nonpolar) Covalent (Equal sharing).
- **0.4 – 2.0:** Polar Covalent (Unequal sharing).
- **> 2.0:** Ionic (Electron transfer).

3.2 10.8: Formal Charge

Used to identify the **best** Lewis structure among resonance forms.

Formula:

$$\text{FC} = (\text{Valence } e^-) - [\text{Unpaired } e^- + \frac{1}{2}(\text{Bonding } e^-)]$$

Selection Rules:

1. Smaller FCs are better (closer to zero).
2. Negative FCs should be on the more electronegative atom.

3.3 10.10: Calculating Bond Energy (ΔH_{rxn})

Meticulous Method: Step-by-Step ΔH_{rxn}

Formula: $\Delta H_{\text{rxn}} = \sum(\text{Bonds Broken}) - \sum(\text{Bonds Formed})$

1. **Draw:** Sketch the Lewis structures for **every** reactant and product.
2. **Inventory Reactants (Breaking):** Count every bond in the reactants. Multiply by the coefficients from the balanced equation. These values are **Positive**.
3. **Inventory Products (Forming):** Count every bond in the products. Multiply by the coefficients. These values are **Negative** in the formula.
4. **Calculate:** [Sum of Reactant Bond Energies] – [Sum of Product Bond Energies].

Note: Breaking bonds costs energy (+), forming bonds releases energy (-).

4 Chapter 11: Molecular Shapes (VSEPR)

4.1 11.2–11.4: VSEPR Theory

Concept: Electron groups (lone pairs and bonds) repel each other and want to be as far apart as possible.

- **Electron Geometry:** Based on total electron groups.
- **Molecular Geometry:** Based only on where atoms are (lone pairs change the name).

Total Groups	Lone Pairs	Electron Geo	Molecular Geo	Ideal Angle
2	0	Linear	Linear	180°
3	0	Trigonal Planar	Trigonal Planar	120°
3	1	Trigonal Planar	Bent	$< 120^\circ$
4	0	Tetrahedral	Tetrahedral	109.5°
4	1	Tetrahedral	Trigonal Pyramidal	$< 109.5^\circ$
4	2	Tetrahedral	Bent	$<< 109.5^\circ$
5	0	Trig. Bipyramidal	Trig. Bipyramidal	$90^\circ, 120^\circ$
6	0	Octahedral	Octahedral	90°

4.2 11.3: Lone Pair Effects

Concept: Lone pairs are "physically larger" than bonding pairs because they are held by only one nucleus. They exert more repulsion, which **squeezes** the bonding angles to be smaller than the ideal values.

5 Summary Checklist

Can I convert units for de Broglie ($g \rightarrow kg$)?

Do I remember that l values 0, 1, 2, 3 correspond to s, p, d, f ?

Can I rank atoms by size: Fr (largest) vs F (smallest)?

Do I know that Z_{eff} increases to the **right**?

Can I calculate Formal Charge for a molecule like N_2O ?

Am I remembering to subtract "Formed" from "Broken" for ΔH ?